University of Rajshahi

Department of Computer Science and Engineering

CSE4261: Neural Network and Deep Learning

Class Test

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Ans. to the ques. no-1.

Here, G stands for Generator in Generative Adversarial Network. At the testing phase we use the trained generator for generating output.

1. Unconditional GAN (UGAN): Takes random noise vector as input ($z \sim N(0,1)$) and generate synthetic image from the learned distribution. Generate realistic sample but without control. Example: Trained on MNIST; at test G generates digit, but we don't know if its 3, 7, or 9 until we see it.

2. Condition GAN (cGAN): Takes z + condition as input and generates samples that respect the condition. e.g. If we say condition = "7" than G will generate image that look like 7.

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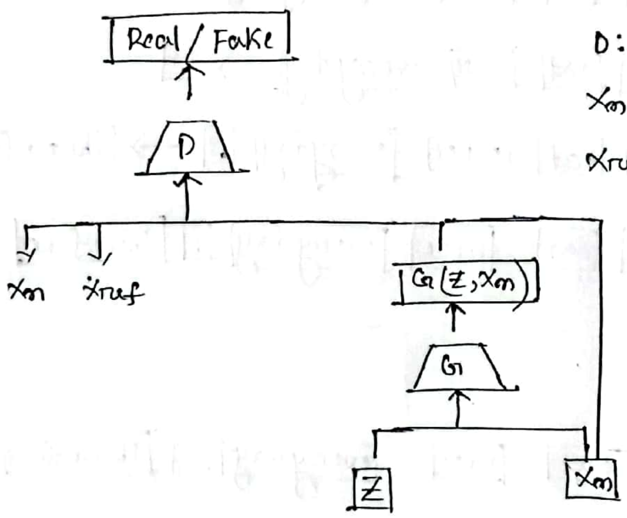
Ans. to the ques. no-2

As given a neutral face and a reference face, I can use them as a pair to generate the expression. Neutral + reference expression image makes a training pair. So to deal with paired data I will use conditional GAN (cGAN).



G: Generator

D: Discriminator

X_n: Neutral Image

X_{ref}: Reference expression image

$G(Z, X_n)$: fake expression image

Z: Random Noise

Fig: cGAN Design for expression generation

A cGAN has two network. The Generator and Discriminator. Generator will take a noise vector and the neutral face as

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conditional input and produce a synthetic expression image while preserving the candidate identity. The discriminator then evaluate two pairs: one combining the neutral image with the real expression, and another combining the neutral image and generated expression in order to distinguish real from fake.

multiple cGAN could be trained for different expressions (smile, anger, etc). This strategy solve the single sample per person (SSPP) problem by generating multiple synthetic expression from one neutral image.

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Ans. to the question no — 3

Xception: Based on pointwise + depthwise ~~seperable~~ convolution which corresponds to depthwise seperable convolution.

VGG16: uses very deep architecture with stacked 3x3 convolutional kernels instead of large kernel showing depth with small kernel can significantly improve performance.

ResNet50: Introduces residual (skip) connections, which solve the vanishing gradient problem.

InceptionResNetV2: combines multi scale feature extraction of inception modules residual connections, achieving efficiency and stable training.

MobileNet: Uses depthwise separable convolution which drastically reduces the numbers of parameter.

DenseNet121: Each layer connect all previous layer, enabling effiecient feature reuse and strong gradient flow.

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Ans. to the ques. no – 4

The AI recruitment assistant of Amazon suffered from dataset bias. Since the training dataset was highly imbalance with mostly male resumes, the deep learning model's learned feature representations became skewed towards pattern associated with male applicants. consequently during inference resumes containing female related features were treated as outliers and assigned lower probability. This resulted in biased decision boundaries.

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Ans. to the ques. no - 5

Derivation of

KL Divergence = cross entropy - Entropy.

L.H.S: we know,

$$D_{KL}(P \| Q) = \sum_i P(x_i) \times \log \frac{P(x_i)}{Q(x_i)}$$; where P is true

Distribution and Q is predicted distribution.

R.H.S: Formula for cross entropy $H(P, Q) = -\sum_i P(x_i) \log(Q(x_i))$

and Entropy $H(P) = -\sum_i P(x_i) \log(P(x_i))$

Now cross Entropy - Entropy

$$= H(P, Q) - H(P)$$

$$= -\sum_i P(x_i) \log(Q(x_i)) + \sum_i P(x_i) \log(P(x_i))$$

$$= \sum_i P(x_i) \{ \log(P(x_i)) - \log(Q(x_i)) \}$$

$$= \sum_i P(x_i) \log \frac{P(x_i)}{Q(x_i)}$$

$$= D_{KL}(P \| Q).$$

This KL Divergence tells how much information is lost when using distribution 'Q' to approximate distribution 'P'.

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Ans. to the ques no - 6

Type of model at E_1, E_2, E_3,

(i) E_1: Early stopped, both training and validation loss still high. The model is underfitted. It hasn't learned enough patterns from the data yet.

(ii) E_2: Training loss and validation loss both are reasonably low and model is optimal fit at this stage.

(iii) E_3: Training loss keeps decreasing but validation loss increased, the model is overfitted. It memorized training data but lost generalization on unseen data.

Now, how to stop training at right epochs:

1. Early stopping: We can use regularization technic where training is automatically stopped when the validation loss stops improving for several epoch ensuring the best model is saved.

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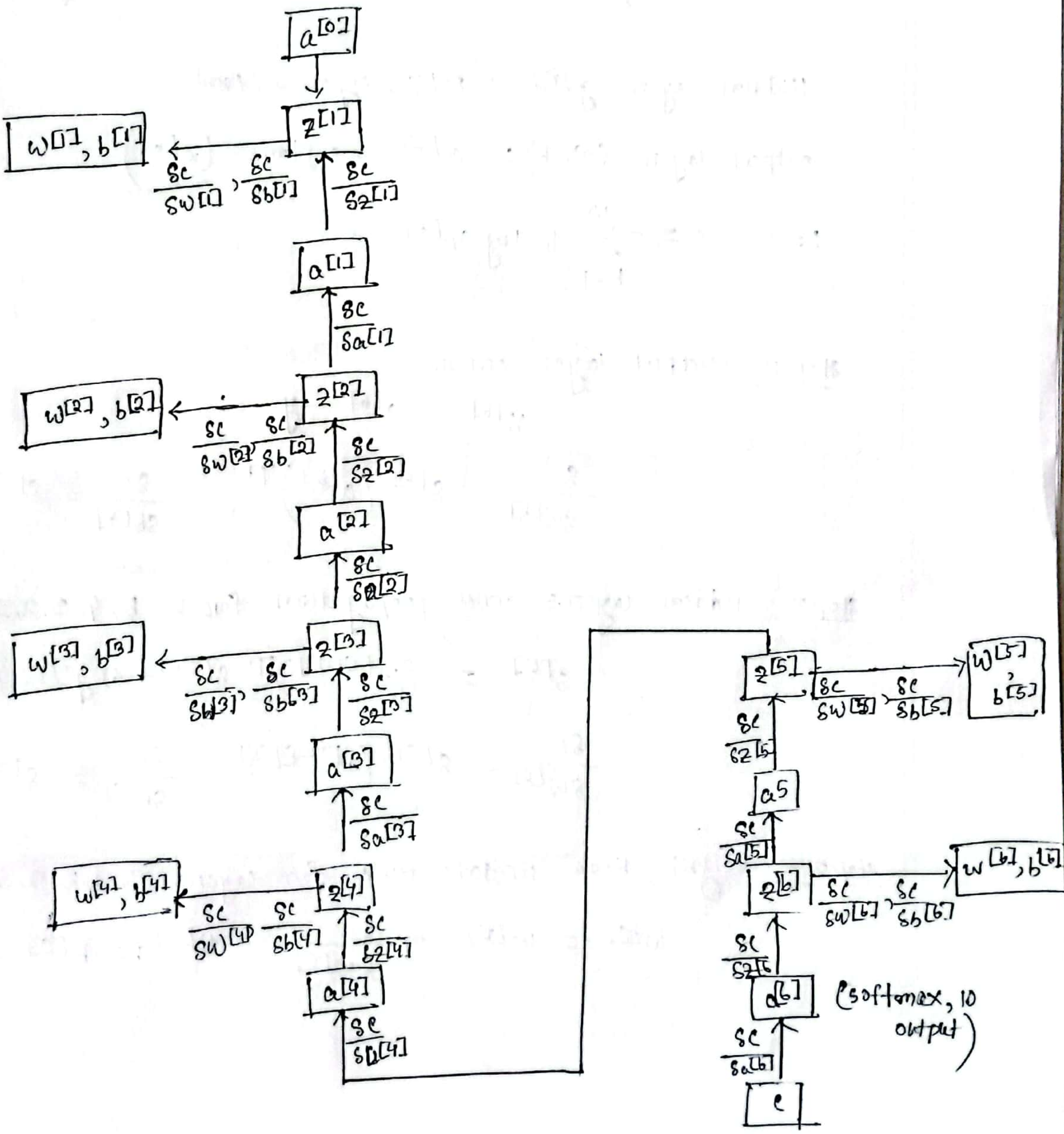
Ans. to the ques. no-7

Back propagation computation graph for neural network having 5 hidden layers for classifying images of 10 English digit.



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Equation for updating weight by back propagation algorithm:

Forward pass:

For each layer $\ell = 1, 2, 3, 4, 5, 6$;

$$z^{[\ell]} = w^{[\ell]} a^{[\ell-1]} + b^{[\ell]}, \quad a^{[\ell]} = g^{[\ell]}\left(z^{[\ell]}\right)$$

Hidden layer $g^{[\ell]}$ = ReLU / sigmoid / tanh

output layer $(\ell = 6)$: $a^{[6]} = \text{softmax}\left(z^{[6]}\right)$

Loss $\quad C = -\sum_{k=1}^{10} y_k \log a_k^{[6]}$

Backward pass:

Step 1: output layer error

$$\delta^{[6]} = a^{[6]} - y$$

$$\frac{\delta C}{\delta w^{[6]}} = \delta^{[6]}\left(a^{[5]}\right)^T, \quad \frac{\delta C}{\delta b^{[6]}} = \delta^{[6]}$$

Step 2: Hidden layers error propagation for $\ell = 5, 4, 3, 2, 1$

$$\delta^{[\ell]} = \left(w^{[\ell+1]}\right)^T \delta^{[\ell+1]} \odot g'^{[\ell]}\left(z^{[\ell]}\right)$$

$$\frac{\delta C}{\delta w^{[\ell]}} = \delta^{[\ell]}\left(a^{[\ell-1]}\right)^T, \quad \frac{\delta C}{\delta b^{[\ell]}} = \delta^{[\ell]}$$

Step 3: weight, bias update rule for layers $\ell = 1, 2, 3, 4, 5, 6$

$$w^{[\ell]} := w^{[\ell]} - \eta \frac{\delta C}{\delta w^{[\ell]}}, \quad b^{[\ell]} := b^{[\ell]} - \eta \frac{\delta C}{\delta b^{[\ell]}}$$

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Ans. to the ques. no-8

BERT uses masked language modeling (MLM) and next sentence prediction (NSP) for self supervised pre-training.

Mask Language model (MLM): Randomly mask some tokens in input sentence and let the model predict them.

Given text: "I eat rice everyday. Therefore, I need to cook rice".

input vector:

$[CLS]\ \boxed{I}\ .\ \boxed{eat}\ ,\ \overset{[mask]}{\boxed{rice}}\ \boxed{everyday}\ .\ \boxed{[SEP]}\ \boxed{Therefore}\ \boxed{,}\ \boxed{I}$

$\boxed{need}\ \boxed{to}\ \overset{[mask]}{\boxed{cook}}\ \boxed{rice}\ \boxed{.}\ \boxed{[SEP]}$

Next sentence Prediction (NSP):

sentence A: "I eat rice everyday".

sentence B: "Therefore, I need to cook rice."

Label = isNext

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GPT (Generative Pretrained Transformer):

Causal Language modeling (CLM):

input → output

I → eat

I eat → rice

I eat rice → everyday

I eat rice everyday → .

I eat rice everyday . → Therefore

I eat rice everyday . Therefore → ,

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I eat rice everyday . Therefore , I need to cook rice

→ .

This is how sequential token prediction is repeated for all the token.

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Ans. to the ques no - 9

Given, feature map:

$$\begin{bmatrix} 3 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 2 \\ 0 & 0 & 2 & 2 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

channel 1 channel-2

Kernel :

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

channel 1 channel 2

(i) Regular convolution without zero padding and strides = 2.

Dimension of the output $= \left[\frac{n - k + 2P}{s}\right] + 1$

$= \left(\frac{4 - 2 + 2\times 0}{2}\right) + 1 = 2$

output will be 2x2 matrix

where
P = Padding
s = stride
k = kernel
n = input

channel will merge to single channel. Stride 2 slide over 2 pixel at a time.

$$\begin{bmatrix} 3 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 2 \\ 0 & 0 & 2 & 2 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 9 & 6 \\ 7 & 12 \end{bmatrix}$$

output

(ii) Regular convolution with zero padding and stride 2:

$$\text{output Dimension} = \left(\frac{n-k+2P}{s}\right)+1$$

$$= \left(\frac{4-2+2.1}{2}\right)+1$$

$$= 3$$

output will be ~~4×4~~ 3×3

$$\begin{vmatrix} \begin{bmatrix}0&0\\0&3\end{bmatrix} & \begin{bmatrix}0&0\\2&1\end{bmatrix} & \begin{bmatrix}0&0\\1&0\end{bmatrix} \\ \begin{bmatrix}0&0\\0&1\end{bmatrix} & \begin{bmatrix}1&1\\2&0\end{bmatrix} & \begin{bmatrix}0&0\\2&0\end{bmatrix} \\ \begin{bmatrix}0&0\\0&0\end{bmatrix} & \begin{bmatrix}0&2\\0&0\end{bmatrix} & \begin{bmatrix}2&0\\0&0\end{bmatrix} \end{vmatrix} \quad \begin{vmatrix} \begin{bmatrix}0&0\\0&1\end{bmatrix} & \begin{bmatrix}0&0\\1&1\end{bmatrix} & \begin{bmatrix}0&0\\1&0\end{bmatrix} \\ \begin{bmatrix}0&0\\0&2\end{bmatrix} & \begin{bmatrix}0&0\\2&2\end{bmatrix} & \begin{bmatrix}0&0\\2&0\end{bmatrix} \\ \begin{bmatrix}0&0\\0&0\end{bmatrix} & \begin{bmatrix}0&0\\0&0\end{bmatrix} & \begin{bmatrix}0&0\\0&0\end{bmatrix} \end{vmatrix} \otimes \begin{vmatrix}1&2\\2&1\end{vmatrix} \begin{vmatrix}1&0\\0&1\end{vmatrix}$$

=

4	6	2
3	9	4
0	4	2

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(iii) Depthwise convolution without zero padding, with stride = 1.

$$\text{output Dimension} = \left(\frac{n-k+2P}{s}\right)+1$$

$$= \left(\frac{4-2+0}{1}\right)+1$$

$$= 3$$

output = 3x3

Depthwise convolution do not mix channel. Each channel is convolued with separate kernel. Output will be two channel.

$$\begin{vmatrix} 3 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 2 \\ 0 & 0 & 2 & 2 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{vmatrix} \otimes \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

⇒

8	7	5
6	7	3
5	4	10

channel 1

1	1	1
2	2	2
2	2	2

channel 2